Class Test
Subject: Fluid Mechanics
3^{rd} Semester CE (Sec.A)

Date: 10.11.2025 **Total Marks: 30**

Q.1 State Bernoulli's theorem for steady flow of an incompressible fluid. Derive an expression for Bernoulli's theorem and state the assumptions made for such a derivation. [8+2=10]

Q.2 The velocity vector in a fluid flow is given by $\vec{V} = 2x^3\hat{i} - 5x^2 y\hat{j} + 4t\hat{k}$. Find the velocity and acceleration of a fluid particle at (1, 2, 3) at time $t = 1$. [10]

Q.3 (a) A 25 cm diameter pipe carries oil of sp.gr. 0.9 at a velocity of 3 m/s. At another section the diameter is 20 cm. Find the velocity at this section and also mass rate of flow of oil. [05]

(b) Water is flowing through a pipe of 100 mm diameter under a pressure of 19.62 N/cm^2 (gauge) and with mean velocity of 3 m/s. Find the total head of the water at a cross-section, which is 8 m above the datum line. [05]